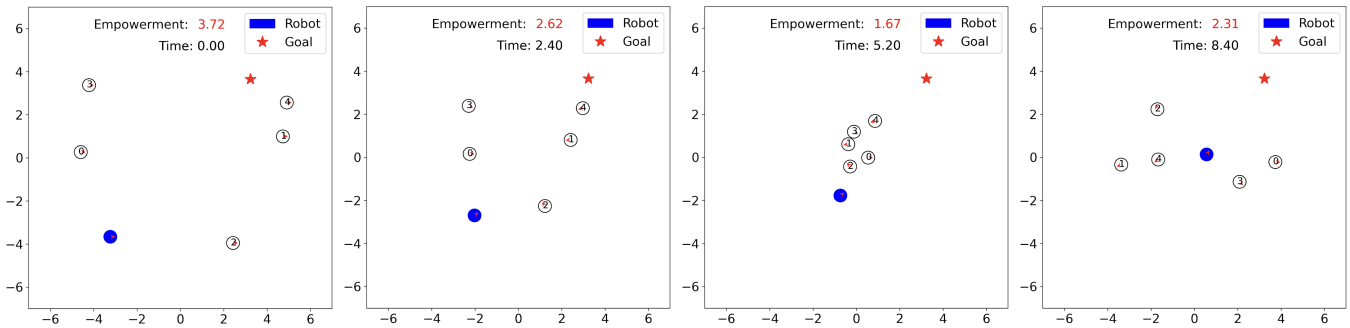

(a) t = 0s (b) t = 6s (c) t = 12s (d) t = 18s

Fig. 4: Time-evolution of the robot's trajectory and corresponding empowerment values at different time steps. The robot (blue circle) navigates towards the goal (red star) while interacting with nearby humans (numbered circles). Empowerment, shown in the top of each subplot, decreases as the robot progresses through the environment, reflecting reduced autonomy due to increased proximity to humans. This figure demonstrates how empowerment evolves dynamically in response to the robot's movement through a crowded environment.

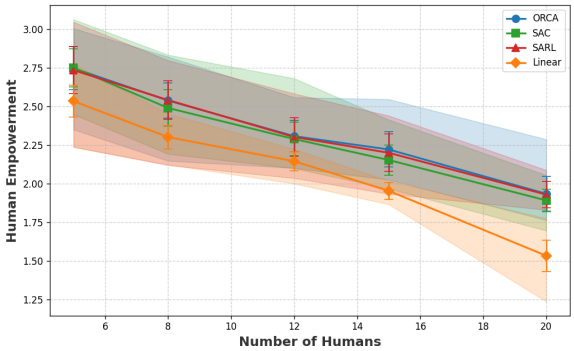

Fig. 5: Avg. Empowerment for Different Policies vs. Crowd Sizes

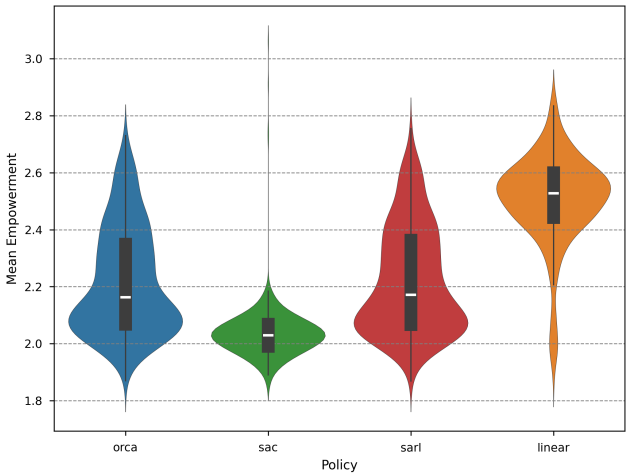

Fig. 6: Violin plots of mean empowerment values across 500 episodes for all policies, showing the distribution and variability for comparison.

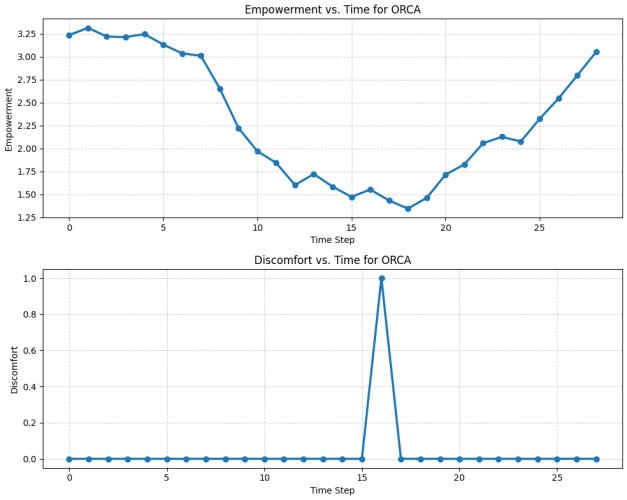

Fig. 7: Comparison of Empowerment against Discomfort Rate

D. Temporal Comparison Between Empowerment and Discomfort Metrics

The two plots in Fig. 7 depict empowerment and discomfort over time for the ORCA policy over a single episode, showing an interesting relationship between the two metrics. As the top graph indicates, human empowerment decreases steadily as the agents get closer to each other, reflecting their diminishing control and flexibility as they navigate through increasingly constrained conditions. Empowerment then begins to recover, rising after some time as the humans regain autonomy.

In contrast, the discomfort metric (bottom graph) remains at zero for most of the timeline but spikes sharply when the humans are close, triggering a discomfort response. Clearly, this binary metric is useful only to indicate whether or not a critical event has occurred (e.g., a close encounter) without providing a sense of the gradual changes leading up to that event. Human empowerment on the other hand, as a contin-

uous variable, offers the benefit of reflecting a more nuanced picture, as it continuously tracks a human's ability to navigate freely. This makes empowerment a more informative metric for capturing the dynamics of human-robot interactions.

VI. Statistical Analysis

To validate our proposal of using human empowerment as a metric for measuring the performance of a robot's policy, we demonstrate its ability to differentiate the implicit influence of various policies. In this section, we present our statistical analysis to support this objective.

We conduct 500 trials for each of four different policies, SARL, SAC, ORCA, and Linear. The objective is to evaluate and compare the impact of these policies on human empowerment. For each trial, we calculate the Mean Empowerment, a key metric for our analysis, and subsequently perform appropriate statistical tests to validate our hypothesis.

In each trial, we compute human empowerment for every individual human at each time step, starting from the beginning of the scenario until the human reached their destination. Then, we compute the average empowerment of each human throughout their trajectory to capture their empowerment throughout the interaction. To derive the final metric, we calculate the average of these empowerment values across all humans in the scene, which we refer to as **Mean Empowerment** throughout this section. For our statistical analysis, we construct a distribution of 500 random trials for each policy and record the Mean Empowerment parameter. In each trial, we randomize the human initial position based on random seed that the learning-based policy have not seen during their training.

We first assess the normality of the collected data using the **Shapiro-Wilk test** [29] due to our sample size (500 samples per each policy) and its ability to detect deviations in both skewness and kurtosis. The null-hypothesis for this test is that the data follows a normal distribution. Our result (Table II) shows that the mean of average human empowerment is not normally distributed ($p < 0.05$), leading us to reject the null-hypothesis.

TABLE II: Statistical Results for Shapiro-Wilk test

	ORCA	SAC	SARL	Linear
Statistic	0.940	0.632	0.942	0.911
p-value	$2.5e-13$	$2.4e-31$	$5.2e-13$	$1.5e-16$

Further, we use a non-parametric **Kruskal-Wallis test** [30] to determine whether there were statistically significant differences in the empowerment values across the different policies. The null hypothesis for this test states that the distribution of empowerment values is the same across all policies. The test results for this test (Statistics=1233.4, p= $8.9e-266$) rejects the null hypothesis and reveals that **at least one of the policies** has a significantly different distribution compared to the others. Following the Kruskal-Wallis test, we perform **a Dunn's post-hoc test** [31] with Boneferroni correction to identify which specific policy pairs had significant difference between the two policies. The results (Fig. 8) show that mean empowerment for SAC and Linear policies are significantly different (p-value < 0.05) from all the other policies. Clearly, the mean human empowerment metric is capable of distinguishing between the results of different robot navigation policies.

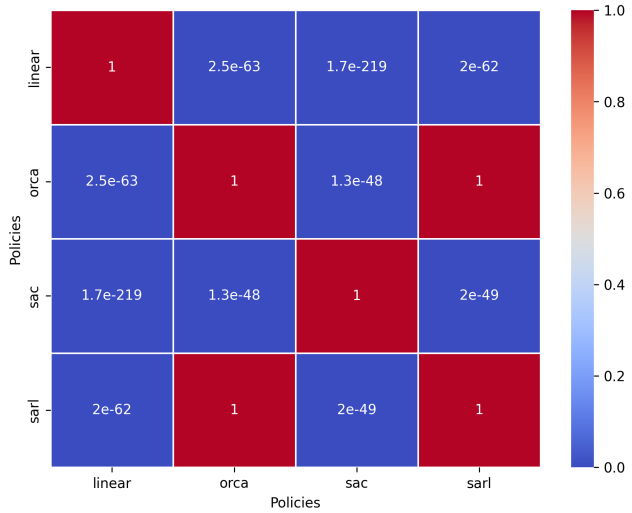

Fig. 8: Dunn's post-hoc test showing the **p-value** between policy pairs.

VII. Conclusion

In this paper, we introduced human empowerment as a novel metric for evaluating social compliance in crowd navigation, addressing the limitations of traditional metrics like proxemics. Our framework offers insights into how the metric captures a human's perceived sense of agency while navigating in shared spaces, and our simulations reveal statistically significant differences across various robot navigation policies. These findings suggest that human empowerment complements the assessment of how robot navigation policies affect human autonomy in dynamic environments, contributing to the broader understanding of human-robot interaction.

However, we acknowledge that this work is an initial exploration of empowerment as a metric for social compliance. Future research should involve more thorough investigations and user studies to assess whether empowerment-based metrics effectively capture human perceptions of socially compliant robot behavior. Additionally, exploring other forms of empowerment, such as transfer empowerment or robot empowerment, could provide deeper insights into human-robot interactions in social navigation contexts.

Moreover, we believe empowerment-based metrics have potential beyond evaluation and could be used in the behavior design for social navigation. By incorporating these metrics into the design process, future systems could better balance autonomy with social compliance, enhancing the quality of human-robot interaction in complex, shared environments.

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